

EVAN KERR

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Education

McMaster University

September 2021 – December 2026

Bachelor of Engineering in Mechanical Engineering

Hamilton, ON

- **cGPA:** 10.7/12.0
- **Awards:** Senate Scholar - 2025/26 (Top 1% Fall-Winter Average), Dean's Honours List
- **Coursework:** Machine Design, CAD/FEA, Materials Science, Thermo-Fluids Systems Design, Manufacturing

Technical Skills

Design & Analysis: Solid Edge, Inventor, FEA (ANSYS), AutoCAD, GD&T (ASME Y14.5), DFMA, Teamcenter

Manufacturing: Machining (Lathe, Milling, Drill Press), Welding, Rapid Prototyping, 3D-Printing, RCA, DOE

Languages/Tools: Python, MATLAB, MS Office (Excel, Powerpoint, Word, Project), MathCAD, PowerBI, Bluebeam

Experience

Hatch

May 2024 – August 2025

Mechanical Engineering Intern

Mississauga, ON, In-Person

- Designed & assembled sensor **fixture & enclosure sheet metal** equipment for **Machine Vision** applications in **Autodesk Inventor**. Verified **DFM** compliance for **10+** sheet metal parts with vendors.
- Contributed to **ASME B30** - compliant furnace **hoist weldment** detailed design work in **Solid Edge** and **Teamcenter**. Verified weldment structural integrity in **ANSYS**, identifying and re-designing weldment components to **reduce** critical stresses by **35%**.
- Drafted multi-phase **steel plant layouts** in **AutoCAD** for a process feasibility study. Integrated design feedback through weekly **design reviews**, as well as weeklong workshops on site with **30+ international stakeholders**.
- Developed a **Python**-based probabilistic corrosion model based on **NDT** inspections, streamlining data entry to reduce setup time by **74%**. Validated statistical accuracy in **MathCAD**.
- Contributed to the development of steel fabrication standard **specifications**, saving **\$5k** in project costs per year.

Ericsson

May 2023 – August 2023

Project Coordinator Intern

Mississauga, ON, Hybrid

- Optimized **Project Controls** workflows for **KPI** dashboards & **QA** documentation, directly supporting Engineering Managers in tracking deliverables and maintaining audit-readiness.
- Managed data organization for **70+** ongoing project plans under **20+** PMs using **Microsoft Project and Excel**, reducing reporting errors by **95%**.
- Generated modified views of **15+** ongoing project timelines using **Excel** and **Microsoft Timeline**.

Projects

Conforming Pneumatic End Effector

- Collaborated with **Honda** to design an adaptive **pneumatic** end effector for automotive assembly, achieving a **10kg** payload capacity and a **75mm** grip stroke for varied component geometries.
- Re-engineered assembly equipment using **ANSYS & Inventor** to validate structural safety factors, increasing lifting capacity by **80%** while eliminating critical **failure modes**.

Wind Tunnel Turbulence Intensity Characterization

- Utilized **DOE** principles to characterize **turbulence intensity (TI)** as a function of fan speed and test section vertical depth, resulting in nominal TI ranges of **1.15% - 2.4%**. Utilized **MATLAB** for data filtration & visualization.

Extracurriculars

FUSE, McMaster University

September 2025 – Present

Teaching Assistant - Engineering Design

McMaster University

- Assisting **40+** students in weekly design labs, applying **Autodesk Inventor** to develop students' **2D/3D CAD** skills.
- Preparing lab material & grading assignments on a **weekly** basis, **reinforcing topics** in engineering design and providing **tangible feedback** to improve technical skills.

McMaster Society of Mechanical Engineers

September 2023 – Present

Year Representative

McMaster University

- Representing **800+** mechanical engineering students in bi-weekly meetings, communicating feedback to faculty and contributing to measurable course delivery improvements.
- Contributing to the execution of **5+** departmental events per term, boosting student engagement within the faculty.